

Problem 2

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Problem 1 Let $f(x)$ and $g(x)$ be functions for which we only know the following:

$$\int_1^4 f(x) dx = 7 \quad \int_2^4 f(x) dx = 5 \quad \int_1^4 g(x) dx = 2$$

Compute the following integrals, if possible. If it is not possible, give examples explaining why not.

(a) $\int_1^4 (8f(x) - 7g(x)) dx$

(b) $\int_1^2 (-f(x)) dx$

(c) $\int_1^4 |f(x)| dx$

(d) $\int_1^4 (2 - x + f(x)) dx$